

Daily Practice Problem (DPP)

Topic: Propositions, Implication, Biconditional, Truth tables, Predicates and Quantifiers

Name: _____

- Suppose that x and y are two Boolean variables in a programming language where True and False are 1 and 0. (That is, the values of x and y are both 0 or 1.) Furthermore, in many languages (including Python), anything that is not False (in other words, anything other than 0) acts like True for the purposes of conditionals; for example, `if 2 then X else Y` will run successfully and execute X. Each of the following code snippets includes a conditional statement based on an arithmetic expression using x and y . Rewrite the given condition using the standard notation of propositional logic.
 - `if x * y ...`
 - `if x + y ...`
 - `if 2 - x - y ...`
 - `if (x * (1 - y)) + ((1 - x) * y) ...`
 - Assume that $\{p_1, \dots, p_n\}$ are all Boolean variables in Python — that is, their values are all 0 or 1. Write a Python conditional expressing the condition that at least 3 of $\{p_1, \dots, p_n\}$ are true.

- In addition to purely logical operations, computer circuitry has to be built to do simple arithmetic very quickly. Consider a number

$$x \in \{0, \dots, 15\}$$

represented as a 4-bit binary number, and denote by x_0 the least-significant bit of x , by x_1 the next bit, and so forth. (Think of 0 as false and 1 as true.) Here you'll explore some pieces of using propositional logic and binary representation of integers to express arithmetic operations. (It's straightforward to convert your answers into circuits.) See Figure below.

x_3	x_2	x_1	x_0		x_3	x_2	x_1	x_0		x_3	x_2	x_1	x_0	
0	0	0	1		0	0	1	1		1	1	0	0	
				$0 + 0 + 0 + 1 = 1$					$0 + 0 + 2 + 1 = 3$					$8 + 4 + 0 + 0 = 12$

Fig (a). Representing small numbers with four Boolean variables (where 0 = False and 1 = True).

- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is greater than or equal to 8.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 4.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 5. (Hint: build a truth table.)
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is evenly divisible by 9.
- Give a proposition over $\{x_0, x_1, x_2, x_3\}$ that expresses that x is an exact power of two.
- Suppose that we have two 4-bit input integers x and y , represented as in questions a–e. Give a proposition over the Boolean variables $\{x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3\}$ that expresses the condition that $x = y$.
- Give a proposition over $\{x_0, x_1, x_2, x_3, y_0, y_1, y_2, y_3\}$ that expresses the condition $x = 2 \cdot y$.

3. Translate each of these statements into logical expressions in three different ways by varying the domain and by using predicates with one and with two variables.
 - (a) A student in your school has lived in Vietnam.
 - (b) There is a student in your school who cannot speak Hindi.
 - (c) A student in your school knows Java, Prolog, and C++.
 - (d) Everyone in your class enjoys Thai food.
 - (e) Someone in your class does not play hockey.

4. Translate these specifications into English where $F(p)$ is "Printer p is out of service," $B(p)$ is "Printer p is busy," $L(j)$ is "Print job j is lost," and $Q(j)$ is "Print job j is queued."
 - (a) $\exists p (F(p) \wedge B(p)) \rightarrow \exists j L(j)$
 - (b) $\forall p B(p) \rightarrow \exists j Q(j)$
 - (c) $\exists j (Q(j) \wedge L(j)) \rightarrow \exists p F(p)$
 - (d) $(\forall p B(p) \wedge \forall j Q(j)) \rightarrow \exists j L(j)$

5. Let $L(x, y)$ be the statement " x loves y " where the domain for both x and y consists of all people in the world. Use quantifiers to express "Joy is loved by everyone".

6. Consider these statements, of which the first three are premises and the fourth is a valid conclusion.

"All hummingbirds are richly colored."

"No large birds live on honey."

"Birds that do not live on honey are dull in color."

"Hummingbirds are small."

Let $P(x)$, $Q(x)$, $R(x)$, and $S(x)$ be the statements " x is a hummingbird," " x is large," " x lives on honey," and " x is richly colored," respectively. Assuming that the domain consists of all birds, express the statements in the argument using quantifiers and $P(x)$, $Q(x)$, $R(x)$, and $S(x)$.

7. Suresh says that in every region there is a town where all inhabitants are happy. Ramesh wants to say that Suresh is wrong. Which of the following sentences should Ramesh say?
- (a) There is a region where there is a town where all inhabitants are happy.
 - (b) In every region in all towns all inhabitants are happy.
 - (c) In every region there is a town where at least one inhabitant is unhappy.
 - (d) There is a region where in all towns at least one inhabitant is unhappy.
8. Mohit claims that every student in a group knows Hindi, English, or both. Which of the following sentences asserts that Mohit's statement is wrong (no more, no less)?
- (a) All students know at most one of these two languages.
 - (b) There is a student who does not know Hindi and does not know English.
 - (c) There is a student who does not know Hindi or does not know English (or both).
 - (d) Every student who knows French does not know German.
9. Let $S(x)$ be the predicate "x is a student," $F(x)$ the predicate "x is a faculty member," and $A(x,y)$ the predicate "x has asked y a question," where the domain consists of all people associated with your school. Use quantifiers to express each of the following statements:
- (a) Some students have not asked any faculty member a question.
 - (b) There is a faculty member who has never been asked a question by a student.
 - (c) Some students have never been asked a question by a faculty member.
10. Express the negations of these propositions using quantifiers, and in English.
- (a) Every student in this class likes mathematics.
 - (b) There is a student in this class who has never seen a computer.
 - (c) There is a student in this class who has taken every mathematics course offered at this school.

Connecting Logic With Computer Science

11. Consider the following program, designed to interchange the values of two variables x and y :

```
temp := x
x := y
y := temp
```

Find predicates that we can use as the *precondition* and the *postcondition* to verify the correctness of this program. Then explain how to use them to verify that for all valid inputs the program does what is intended.

12. Here some code that uses conditionals, or compound propositions as conditions given as:

- (a)

1 if $x > 20$ or $(x \leq 20$ and $y < 0)$ then
2 $foo(x, y)$
3 else
4 $bar(x, y)$
- (b)

1 if $y \geq 0$ or $y \leq x$ or $(x - y) \cdot y \geq 0$ then
2 $foo(x, y)$
3 else
4 $bar(x, y)$

- (c)

1 if $x \bmod 12 = 0$ then	
2 if $x \bmod 4 \neq 0$ then	
3 $foo(x, y)$	
4 else	
5 $bar(x, y)$	
6 else	
7 if $x = 17$ then	
8 $baz(x, y)$	
9 else	
10 $quz(x, y)$	
- Recall that $x \bmod k$ is the remainder when dividing x by k . So $x \bmod k = 0$ when $k \mid x$ —that is, when x is evenly divisible by k .

Some code that uses nested conditionals, or compound propositions as conditions.

- a. Simplify the code in Figure.(a) as much as possible. (For example, if $p \Rightarrow q$, it's a waste of time to test whether q holds in a block where p is known to be true.)
 - b. Repeat for the code in Figure b.
 - c. Repeat for the code in Figure c.
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